

EIN3 → Response to Hypoxia

(GO:0001666): Hypothesis Adjudication Report

Gene: EIN3 (ETHYLENE INSENSITIVE 3), *Arabidopsis thaliana* **UniProt:** O24606 (EIN3_ARATH) ·
Locus: At3g20770 **Focus:** function_assignment —
existing_annotations[16].function_hypothesis **Seed hypothesis:** EIN3 has response to hypoxia (GO:0001666) **Annotation under review:** GO:0001666, evidence IMP, reference [PMID:25284079](#), qualifier **acts_upstream_of_or_within**, assigned by TAIR

Summary

Executive verdict — **source-specific: REFUTED / over-annotated. Overall biological: weakly / partially supported (indirect, redundant, upstream).**

The EIN3 → *response to hypoxia* (GO:0001666) IMP annotation does not survive independent reading of its cited source. [PMID:25284079](#) (Xie et al., 2015, *The Plant Journal*) is a study of the acyl-CoA-binding protein **ACBP3**, not of EIN3. In that paper **ein3-1** appears only as one genotype in an epistasis panel built around an ACBP3 overexpressor (OE-1). The decisive experiment (Figure 6b) shows that the **ein3-1** single mutant was **not significantly different from wild type** under either dark submergence (DS) or light submergence (LS), and that the **OE-1 ein3-1** double combination **still resembled the hypoxia-sensitive OE-1 parent**. In other words, removing EIN3 neither produced a hypoxia phenotype on its own nor modified the ACBP3 phenotype. The authors' own conclusion attributes the attenuated hypoxic tolerance of ACBP3 overexpressors to **NPR1- and CTR1-mediated signaling**, explicitly not to EIN3. An IMP (Inferred from Mutant Phenotype) annotation requires a mutant phenotype; here the mutant phenotype was null. The paper's proposed EIL1 redundancy was **invoked to explain the null result but never tested** — no **ein3 eil1** double mutant was assayed for hypoxia in this work. This is a textbook case of a database carry-over annotation where the qualifying phenotype does not exist.

Separately from the source, the *broader biology* offers modest, indirect support for an EIN3/EIL1 role in low-oxygen adaptation, but only as an **upstream ethylene-signaling** step and only in **redundancy with EIL1**. Hartman et al., 2019 (PMID:31488841) show that ethylene perception on submergence pre-adapts plants to hypoxia by depleting nitric oxide (via PHYTOGLOBIN1) and thereby stabilizing group VII ERF transcription factors; this pathway runs through EIN2 and the redundant EIN3/EIL1 pair, and EIN3-GFP nuclear accumulation is used only as a *readout* of ethylene perception. Crucially, direct oxygen sensing and the transcriptional activation of hypoxia-responsive genes are executed by the **N-degron pathway and group VII ERFs** (RAP2.2/2.12/2.3, HRE1/2; PMID:26668304), not by EIN3.

Bottom line for the curator: The specific IMP annotation to

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should be **removed or re-sourced**, because its cited experiment reports a null EIN3 phenotype. If any hypoxia link is to be retained for EIN3, it should be a **non-core, indirect BP** annotation citing Hartman et al. 2019, with a qualifier that does not imply direct oxygen sensing, and it should acknowledge EIN3/EIL1 redundancy. No molecular-function (MF) term implying oxygen sensing is warranted. Identity was verified: O24606 is EIN3 (At3g20770), distinct from its paralog EIL1 (At2g27050), so there is no paralog misassignment — the annotation is on the correct protein but rests on inadequate evidence.

Key Findings

F001 — The cited source

(**P** 25284079)

reports a *null ein3-1* hypoxia phenotype; it does not support the IMP annotation

PMID:25284079 (Xie et al., 2015, *Plant J*, full text PMC4309432) is an **ACBP3** paper — "*Arabidopsis acyl-CoA-binding protein ACBP3 participates in plant response to hypoxia by modulating very-long-chain fatty acid metabolism.*" EIN3 is not the subject of the study. The **ein3-1** mutant enters only as one of several genotypes in an epistasis screen designed to place the ACBP3 overexpressor (OE-1) hypoxia-sensitivity phenotype within a signaling network.

The relevant result is in the Results subsection "Hypoxia hypersensitivity in OE-1 is dependent on NPR1 and CTR1" and Figure 6b. Two verbatim observations define the evidence:

"the `ein3-1` mutant was not significantly different to wild type under either DS or LS (Figure 6b), possibly due to the functional redundancy of the EIL1 gene in Arabidopsis."

"the OE-1 `coi1-2` and OE-1 `ein3-1` double combinations were hypoxia sensitive, resembling the OE-1 phenotype."

Both statements point the same way. First, the `ein3-1` single mutant had **no hypoxia phenotype** (no significant difference from wild type under either dark or light submergence). Second, adding `ein3-1` to the ACP3 overexpressor **did not rescue or modify** the OE-1 hypoxia-sensitivity phenotype — the double still behaved like OE-1. Both are the signatures of a gene that is *not required* for the phenotype under assay. The abstract's own conclusion makes the exclusion explicit:

"we observed that the attenuated hypoxic tolerance in ACP3-OEs was dependent on NPR1- and CTR1-mediated signaling pathways"

EIN3 is not named as a required node; NPR1 and CTR1 are.

Methodologically, the annotation is further weakened by the *quality* of the negative result reported. No survival counts, no sample size (n), and no explicit statistical test are reported for the `ein3-1` comparison — the claim is a qualitative "not significantly different." An IMP (Inferred from Mutant Phenotype, ECO:0000315) annotation is meant to record a phenotype *conferred* by loss of the gene. Here the loss-of-function mutant conferred no detectable phenotype. Using a null result to justify a positive functional annotation inverts the logic of IMP evidence.

F002 — The proposed EIL1 redundancy was invoked, not tested; independent biology supports only an indirect, redundant, upstream role

The paper itself hedges the null `ein3-1` result by "possibly due to the functional redundancy of the EIL1 gene." That is a hypothesis to explain absence of phenotype, **not evidence of function**. No `ein3 ein1` double mutant was assayed for hypoxia tolerance anywhere in

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Redundancy invoked to rescue a null result cannot be used to support the very annotation the null result fails to justify.

However, redundancy of EIN3/EIL1 in hypoxia adaptation is independently supported by a different, stronger study. Hartman et al., 2019 (PMID:31488841, *Nat Commun*) demonstrate that ethylene perception pre-adapts *Arabidopsis* to subsequent hypoxia:

"This ethylene-mediated NO depletion and consequent ERFVII accumulation pre-adapts plants to survive subsequent hypoxia"

Mechanistically, entrapment of ethylene during submergence is perceived through the canonical pathway (EIN2 → EIN3/EIL1), leading to PHYTOGLOBIN1-mediated depletion of nitric oxide, which stabilizes group VII ERF transcription factors (ERF-VIIs) and thereby pre-adapts the plant. In that study the `ein3eil1-1` **double** mutant is among the ethylene-signaling mutants in which the pre-adaptation response is abolished, and EIN3-GFP nuclear accumulation is used as a readout of ethylene perception on submergence. This places EIN3/EIL1 **upstream** of the oxygen-sensing machinery as part of ethylene signal transduction — it is not itself an oxygen sensor or a direct activator of the core hypoxia-responsive gene battery. The requirement is genetically satisfied only by the *double* mutant, consistent with EIN3 and EIL1 being redundant.

Thus the *overall biological* picture allows a defensible but weak, indirect claim: EIN3, redundantly with EIL1, participates in ethylene-mediated pre-adaptation to hypoxia. That claim is *not* the one tested by

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and it is BP-level, indirect, and non-core.

F003 — Identity verified: O24606 = EIN3 (At3g20770); no paralog misassignment, but no oxygen-sensing molecular function

UniProt REST (O24606.json) returns `uniProtKBId = EIN3_ARATH`, protein name "ETHYLENE INSENSITIVE 3", gene `EIN3`, ordered locus At3g20770. The functional keywords are **Activator**, **Chromatin regulator**, **DNA-binding**, **Ethylene signaling pathway**, **Nucleus**, **Transcription regulation** — the profile of a nuclear DNA-binding transcriptional activator in the ethylene pathway. There is **no** oxygen-sensing, dioxygenase, or hypoxia molecular-function keyword.

Among UniProt's GO cross-references, exactly one hypoxia/oxygen-related term appears: GO:0001666 *response to hypoxia*, with evidence code **IMP:TAIR** — i.e., the very annotation under review, and the only hypoxia link on the record.

The paralog **EIL1 is a distinct gene at a distinct locus (At2g27050)**. Therefore the annotation sits on the correct protein — there is no confusion of EIN3 with EIL1 at the identity level. The problem is not misassigned identity; it is inadequate evidence for the assigned process.

F004 — QuickGO confirms a single, TAIR-only annotation with a loose qualifier and no orthology support

The EBI QuickGO annotation API (`geneProductId=024606, goId=GO:0001666`) returns `numberOfHits = 1`. That single annotation has: evidence **IMP (ECO:0000315)**, reference

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`assignedBy = TAIR`, qualifier `acts_upstream_of_or_within`, `withFrom = null`, taxon 3702. There is **no second, independent annotation** of GO:0001666 for EIN3, and no with/from ortholog transfer supporting it. The annotation is therefore a **single-source record** whose entire evidentiary weight depends on the one paper analyzed above — and that paper reports a null.

The `acts_upstream_of_or_within` qualifier is itself telling: it is the loosest causal relationship in the GO relations vocabulary, signaling that even the curator did not assert a direct role. That loose qualifier is consistent with — but does not remedy — the underlying null phenotype.

Mechanistic Model / Interpretation

The core question is whether EIN3 *directly has* the function "response to hypoxia," or whether it is a bystander/upstream node that was captured by an epistasis screen and then over-annotated. The evidence supports the latter. The two pathways involved should be kept sharply distinct:

DIRECT OXYGEN SENSING & HYPOXIA-GENE ACTIVATION (EIN3 NOT a member)

$O_2 / NO \rightarrow$ N-degron pathway (PCO oxidation of Cys) $\rightarrow$ ERF-VII stability
 ERF-VIIs (RAP2.2, RAP2.12, RAP2.3, HRE1, HRE2) $\rightarrow$ ~50 hypoxia-responsive genes (HRGs)
 bind Hypoxia-Responsive Promoter Element (HRPE)
 [PMID:26668304, 41732087]

UPSTREAM ETHYLENE SIGNALING (where EIN3/EIL1 acts – indirectly)

Submergence $\rightarrow$ ethylene entrapment $\rightarrow$ ETR1/CTR1 $\rightarrow$ EIN2
 EIN2 $\rightarrow$ EIN3 / EIL1 (redundant TFs, nucleus)
 EIN3 / EIL1 $\rightarrow$ ERF-VII stabilization
 "pre-adaptation to hypoxia"
 [PMID:31488841 (Hartman 2019)]

Submergence $\rightarrow$ PHYTOGLOBIN1 $\rightarrow$ NO depletion $\rightarrow$ ERF-VII stabilization
 "pre-adaptation to hypoxia"
 [PMID:31488841 (Hartman 2019)]

EIN3's authenticated molecular function (F003) is that of a **nuclear DNA-binding transcriptional activator in the ethylene pathway**. Its documented connection to hypoxia is *upstream and indirect*: as part of ethylene signal transduction that can pre-adapt the plant by feeding into ERF-VII stability. It is **not** a component of the oxygen-sensing N-degron machinery and **not** one of the ERF-VII activators that directly bind hypoxia-responsive promoters.

Mapping this onto GO evidence logic:

Layer	Molecular event	EIN3 role	Correct GO framing
Oxygen sensing	Cys oxidation by PCOs, N-degron proteolysis	none	not applicable to EIN3
Direct HRG activation	ERF-VII binding HRPE motif	none	belongs to RAP2.x / HRE1/2
Upstream ethylene relay	EIN2 $\rightarrow$ EIN3/EIL1 transcription	member, redundant with EIL1	non-core, indirect BP (if any)
Source (P 25284079) test	ein3-1 submergence survival	null phenotype	fails IMP requirement

The single IMP annotation conflates the "upstream relay" layer (weakly, indirectly true) with a direct "response to hypoxia" claim, and sources it to a paper whose only EIN3 experiment produced a null. That is the definition of an over-annotation.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
PMID:25284079 (Xie 2015, <i>Plant J</i>)	Mutant phenotype (epistasis)	Refutes (source-specific)	Does loss of EIN3 (ein3-1) alter hypoxia/submergence tolerance?	ein3-1 not significantly different from WT under DS or LS; OE-1 ein3-1 still hypoxia-sensitive like OE-1; phenotype attributed to NPR1/CTR1	<i>Arabidopsis</i> whole-plant dark/light submergence survival; EIN3 is a secondary genotype; an ACBP study
PMID:31488841 (Hartman 2019, <i>Nat Commun</i>)	Mutant phenotype + mechanism	Qualifies / weak support (overall biology)	Does ethylene signaling (EIN2–EIN3/EIL1) pre-adapt plants to hypoxia?	Ethylene → NO depletion (PGB1) → ERF-VII stabilization pre-adapts to hypoxia; ein3eil1-1 double abolishes response; EIN3-GFP a perception readout	<i>Arabidopsis</i> submergence hypoxia; genetic requirement satisfied by double mutant
PMID:26668304 (Gasch/Bailey-Serres, <i>Plant Cell</i>)	Direct assay (Y1H, ChIP, mutant)	Competing	Which TFs directly activate hypoxia-responsive genes?	RAP2.2/RAP2.12 (with RAP2.3) are the principal redundant ERF-VII activators binding the HRPE motif; HRE1/2 minor	<i>Arabidopsis</i> hypoxia-responsive gene battery
PMID:41732087 (review)	Review/database	Orientation	What executes		<i>Arabidopsis</i> synthesis

Citation	Evidence type	Supports/Refutes/Qualifies	Claim tested	Key finding	Context
			plant oxygen sensing?	N-degron/PCO/ERF-VII axis is the core oxygen-sensing/response module	
UniProt O24606 (EIN3_ARATH)	Database record	Supports identity	Is O24606 EIN3 At3g20770 and does it carry an oxygen-sensing MF?	Nuclear DNA-binding ethylene-pathway activator; only hypoxia link is GO:0001666 IMP:TAIR; no oxygen-sensing MF keyword	Arabidopsis sequence annotation
QuickGO (O24606 × GO:0001666)	Database record	Supports (provenance)	How many/what annotations back this term?	1 hit: IMP/ECO:0000315, P 25284079 TAIR, qualifier acts_upstream_of_or_within , withFrom null	Arabidopsis

GO Curation Implications

Lead (requires curator verification): remove or re-source the GO:0001666 IMP annotation on EIN3 (O24606).

- **Term type:** GO:0001666 *response to hypoxia* is a **Biological Process** term. There is no molecular-function claim in play, and none is warranted — EIN3 has no oxygen-sensing or dioxygenase activity (F003). Do **not** add any MF term implying direct oxygen sensing.
- **Action on the existing annotation:** The IMP evidence code requires a mutant phenotype attributable to the gene. The cited paper's `ein3-1` mutant showed **no phenotype** and did

not modify the ACP3 phenotype (F001). The evidence therefore does not meet the IMP standard. Recommended action: **remove** the annotation, or at minimum **re-source** it.

- **If a hypoxia link is retained for EIN3:** cite Hartman et al. 2019 (PMID:31488841) instead, framed as a **non-core, indirect BP** role in ethylene-mediated pre-adaptation, with an `acts_upstream_of` relationship and explicit recognition of **EIN3/EIL1 redundancy** (ideally supported by the `ein3eil1` double-mutant evidence, not `ein3-1` alone). This is an IGI/IMP-on-double-mutant style claim, not an IMP on the single mutant.
- **Qualifier:** The existing loose `acts_upstream_of_or_within` qualifier already signals a non-direct role; it is consistent with removal or with a re-sourced, indirect BP annotation, but it does not repair the missing phenotype in the current source.
- **Avoid "protein binding":** Not applicable; the informative framing is transcriptional activator in the ethylene signaling pathway (already captured by other EIN3 annotations).

GO decision table

Element	Current	Recommended lead
Term	GO:0001666 (BP)	Remove; or re-source as indirect BP
Evidence	IMP (ECO:0000315)	Not met by source; if retained via Hartman 2019, use double-mutant genetic evidence
Reference	P 25284079	Replace with P 31488841 if retained
Qualifier	<code>acts_upstream_of_or_within</code>	<code>acts_upstream_of</code> (if retained)
With/ From	null	consider EIL1 (redundancy) if retained
Core?	(implied)	Non-core , indirect, redundant

Mechanistic Scope

- **Immediate molecular function of EIN3 (direct):** sequence-specific nuclear DNA-binding transcriptional activator in the ethylene signaling pathway (UniProt keywords: Activator,

DNA-binding, Nucleus, Ethylene signaling). This is the gene product's authenticated core activity.

- **What GO:0001666 asserts:** participation in the *response to hypoxia* biological process. For EIN3 this is at best an **upstream, indirect** contribution via ethylene signal transduction (EIN2 → EIN3/EIL1), feeding into ERF-VII stabilization (Hartman 2019).
- **Downstream / not-direct effects that must not be conflated with EIN3 activity:** the actual oxygen-sensing step (N-degron/PCO), the direct transcriptional activation of the hypoxia-responsive gene battery (ERF-VIIs binding the HRPE motif), and the whole-plant submergence-survival phenotypes assayed in

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(which are driven by ACP3/very-long-chain fatty acid metabolism and NPR1/CTR1 signaling, not EIN3).

- **Effects inferred only from loss of function:** none usable here — the `ein3-1` loss-of-function produced *no* hypoxia phenotype, so no function can be inferred from it.

Conflicts and Alternatives

1. The source is not an EIN3 study.

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is an ACP3 paper; EIN3 enters only as an epistasis control. Database carry-over of a term from a secondary genotype is the most likely origin of this annotation.

2. **Null result mislabeled as IMP.** The mutant phenotype that IMP requires is absent (`ein3-1` = WT under submergence). This is the central conflict with the seed hypothesis.
3. **Redundancy invoked, not demonstrated.** EIL1 redundancy is a rescue narrative for the null in

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the double mutant was not tested there. Genuine redundancy evidence exists elsewhere (Hartman 2019) but for a different, upstream process.

4. **Competing direct actors.** The direct transcriptional response to hypoxia is executed by group VII ERFs (RAP2.2/2.12/2.3, HRE1/2; [PMID:26668304](#)), not EIN3. Attributing "response to hypoxia" to EIN3 risks implying it is one of these direct activators, which it is not.

5. **No paralog/identity artifact.** Identity is clean: O24606 = EIN3 (At3g20770) ≠ EIL1 (At2g27050). The problem is evidentiary, not identity-based.

Limitations and Knowledge Gaps

- **No raw data / no reanalysis possible.**

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reports the `ein3-1` comparison qualitatively ("not significantly different"). No survival counts, sample size, or test statistic are given, and no deposited raw dataset was located, so an independent reproducible reanalysis of the `ein3-1` phenotype could not be performed. The verdict rests on the authors' own reported null and stated conclusions, read independently. **Gap:** the quantitative strength of the null (was the study powered to detect a small EIN3 effect?) is unknown; a properly powered `ein3-1` and `ein3 eil1` submergence assay would resolve it. *Missing evidence is not refutation of the broader biology — only of the specific IMP claim.*

- **Direction of the Hartman 2019 requirement.** The `ein3eil1-1` double abolishes ethylene-mediated pre-adaptation, but the study's primary aim was the ethylene→NO→ERF-VII axis; the quantitative EIN3/EIL1 contribution versus other ethylene-pathway components is not fully partitioned. **Gap:** an `ein3 eil1` hypoxia-tolerance survival curve with ERF-VII target readouts would quantify the contribution.
- **Curation provenance vs. curator intent.** QuickGO confirms a single TAIR IMP with a loose qualifier, but the internal TAIR curation note that led to term assignment was not retrievable. **Gap:** the original TAIR curation rationale, if it cited a figure other than 6b, should be checked before deletion.
- **Single-species scope.** All primary evidence is *Arabidopsis*; ortholog behavior (e.g., rice EIL1/2 under submergence) may differ and is not with/from-linked here.

Discriminating Tests

The following would most efficiently separate "EIN3 directly has response to hypoxia" from "EIN3 is an upstream, redundant, indirect node":

1. **Powered `ein3-1`, `eil1`, and `ein3 ein1` submergence survival assays** (dark and light), with explicit n and statistics, plus reoxygenation recovery. Prediction: single mutants $\approx$ WT; only the double shows reduced tolerance if the pre-adaptation role is real.
2. **ERF-VII stability / hypoxia-marker readout in `ein3 ein1`** (e.g., ADH1, PDC1, SUS4 induction; RAP2.12 protein) under submergence to test whether EIN3/EIL1 loss impairs the ethylene→ERF-VII pre-adaptation relay (as Hartman 2019 predicts) without EIN3 binding the HRGs directly.
3. **ChIP-seq / DAP-seq of EIN3** across hypoxia-responsive promoters to test directly whether EIN3 binds the core HRG battery (expected: it does not, in contrast to ERF-VIIs and their HRPE motif).
4. **Epistasis re-test of ACP3 with `ein3 ein1` double**, since

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only used `ein3-1`; this would test whether redundancy actually masks a contribution to the ACP3 phenotype (expected: still NPR1/CTR1-dependent).

Curation Leads (require curator verification)

Candidate	action:	Remove	GO:0001666	IMP
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from EIN3, **or** re-source it as a non-core indirect BP.

Candidate	reference	snippets	to	verify	in	the	source
(P 25284079):							

- Results, Fig. 6b: *"the ein3-1 mutant was not significantly different to wild type under either DS or LS (Figure 6b), possibly due to the functional redundancy of the EIL1 gene in Arabidopsis."* - Results: *"the OE-1 coi1-2 and OE-1 ein3-1 double combinations were hypoxia sensitive, resembling*

the OE-1 phenotype." - Abstract: "we observed that the attenuated hypoxic tolerance in ACBP3-OEs was dependent on NPR1- and CTR1-mediated signaling pathways" (EIN3 not named as required).

Candidate replacement reference (if a hypoxia link is retained): [PMID:31488841](#) — snippet to verify: "This ethylene-mediated NO depletion and consequent ERFVII accumulation pre-adapts plants to survive subsequent hypoxia." Frame as indirect, redundant (EIN3/EIL1), `acts_upstream_of`, non-core BP.

Candidate competing-context reference (for scope, not for EIN3 annotation): [PMID:26668304](#) — identifies the actual direct hypoxia transcriptional activators (ERF-VIIs), which EIN3 is not.

Suggested curator questions: - Does the TAIR curation note point to any figure other than 6b? If not, the IMP basis is the null. - Should the annotation be transferred to EIN3/EIL1 as a redundant pair with double-mutant evidence rather than to EIN3 alone?

Suggested experiments: the four discriminating tests above, prioritizing the powered `ein3 eil1` submergence survival + ERF-VII readout.

Final Verdicts

• Source-specific GO verdict

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REFUTED / over-annotated. The cited experiment reports a null `ein3-1` phenotype and attributes the hypoxia phenotype to NPR1/CTR1, not EIN3; EIL1 redundancy was invoked but untested. The IMP evidence standard is not met.

- **Overall biological verdict: Weakly / partially supported.** An indirect, redundant, upstream EIN3/EIL1 role in ethylene-mediated pre-adaptation to hypoxia is defensible (Hartman 2019), but EIN3 is not an oxygen sensor and not a direct activator of hypoxia-responsive genes.
- **Narrowest defensible function:** EIN3 acts, redundantly with EIL1, as an upstream ethylene-signaling transcription factor that can *contribute to* pre-adaptation to hypoxia — i.e., `acts_upstream_of` the low-oxygen response, non-core.
- **What would change the verdict:** a properly powered, statistically reported `ein3-1` (or `ein3 eil1`) submergence-survival experiment showing a reproducible loss-of-function

hypoxia phenotype attributable to E1N3 — ideally with direct evidence that E1N3 binds/activates hypoxia-responsive genes. Absent that, the specific IMP annotation remains unsupported by its source.

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